CS & IT ENGINEERING

Computer Organization Architecture

Memory Organization

DPP- 01

Discussion Notes



By- Vishvadeep Gothi sir

[MCQ]

#Q. The memory cycle time of a memory is 500nsec. The maximum rate with which the memory can be accessed?

Note: Consider memory as byte addressable.

A 500 Bytes / Sec

B 2000 Bytes / Sec

C 2 Mbytes / Sec

D 2 GBytes / Sec

in 500 nsec, data transferred = 1 B

1 ns, —— " —— = $\frac{1B}{500\,ns}$

1 sec, —— " —— = $\frac{1\,B}{500 * 10^{-9}\,sec}$

$= \frac{10^9\,B}{500\,sec}$

$= \frac{1000}{500} * 10^6\,B/sec$

$= 2\,MB/sec$

[NAT]

#Q. The address bus width of a memory of size 4096×8 bits is ___ bits?

no. of cells = 4096 = 2^{12}

add. size = $(\log_2 2^{12})$ bits

= 12 bits Ans.

[NAT]

#Q. Consider a byte addressable memory which has 0.2 GBPS writing rate. The memory access time is __ nanoseconds?

0.2 GB, access time = 1 sec

1 B, —||— $= \frac{1\,sec * 1B}{0.2\,GB}$

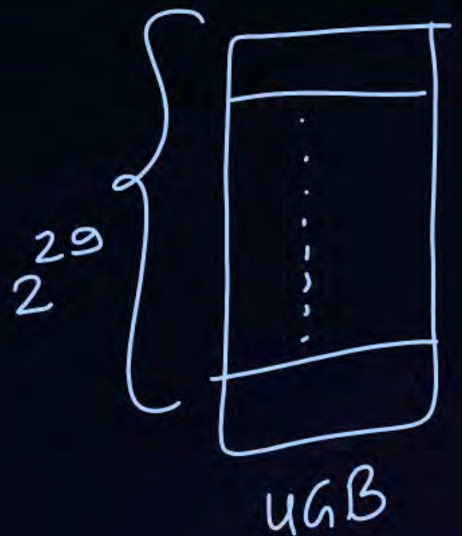
$= 5\,nsec$ Ans

[NAT]

Ans = 8B

#Q. Consider a word addressable memory of total capacity of 4GB. The memory is accessed using a minimum of 29 bits address bus. The word size per address in this memory is ___ bytes?

no. of cells $= 2^{29}$



$$\text{no. of cells} = \frac{4GB}{1 \text{ word size}}$$

$$1 \text{ word size} = \frac{4GB}{2^{29}} = \frac{2^2 \cdot 2^{30}}{2^{29}} B$$

$$= 2^3 B = 8B$$

[MCQ]

#Q. Consider a memory with maximum size of X bytes. Memory is word addressable with word size of W bytes. The size of the address bus of the processor is at least ____ bits?

A $\text{Log}_2(X/W)$

B $2^{(X/W)}$

C X/W

D $\text{Log}_2(X)$

$$\text{no. of cells in mem.} = \frac{\text{Mem. Size}}{\text{word size}}$$

$$= \frac{X \text{ bytes}}{W \text{ bytes}}$$

$$= \frac{X}{W}$$

$$\text{add. size} = \log_2\left(\frac{X}{W}\right) \text{ bits} = \left(\log_2 X - \log_2 W\right) \text{ bits}$$

[MCQ]

no. of cells

#Q. A DRAM chip of 64M × 16 bits has 128K rows of cells with y cells in each row. If DRAM takes x-ns for 1 refresh then total refresh time of the DRAM is ____ microseconds, if $x = 2 \times \log_2 y$?

no. of rows of cells * 18 nsec

= 128 k * 18 nsec

= 128 * 18 μsec

= 2304 μsec

A 1200

B 2304

C 3202

D 5444



$$x = 2 * \log_2 y$$

$$= 2 * \log_2 2^9$$

$$= 18$$

no. of cells = $128k * y$

$$64M = 128k * y$$

$$y = \frac{64M}{128k} = \frac{2^6 * 2^{20}}{2^7 * 2^{10}} = \frac{2^{26}}{2^{17}} = 2^9$$

[NAT]

Ans = 324

#Q. A 32-bits (4B) wide main memory unit with a capacity of 16GB is built using 128M×8-bits RAM chips. If there are x-horizontal arrangements of chips are there, with y number of chips in each horizontal arrangement then the value of 10x + y is?

no. of addresses in expected mem. = $\frac{16GB}{4B} = 4G = 2^{32}$

expected mem = 4G × 4B = 4G × 32 bits

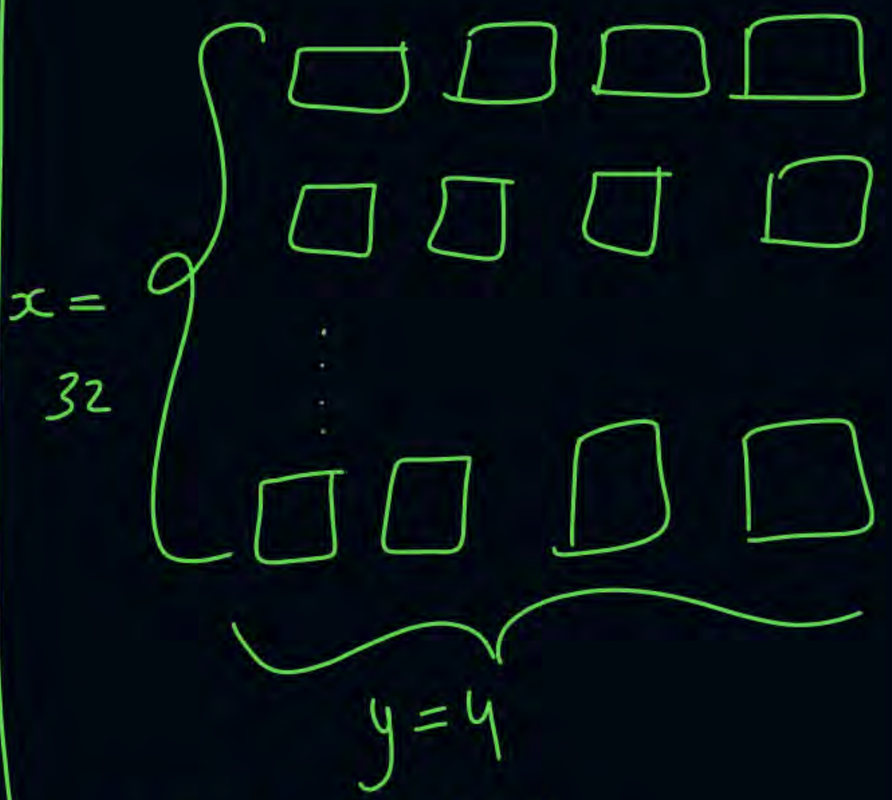
no. of chips $= \dfrac{4G \times 32 \text{ bits}}{128M \times 8 \text{ bits}}$

$= \dfrac{2^5 \times 32 \times 4 \text{ bits}}{2^{27} \times 8 \text{ bits}}$

$= 32 * 4$

$= 128$ chips

arrangement of chips



$10x + y = 10 * 32 + 4$

$= 324$

[NAT]

#Q. Consider a memory system with word addressable memory. The maximum memory capacity is 2Gbytes and memory can be accessed with 28-bits address. The size of word per address is ____ bits?

Memory capacity = 2GB = 2^{31} B

No of addresses = 2^{28} = No of words

1 word size = $\frac{2^{31} B}{2^{28}}$ = 2^3 B = 8B = 8 × 8 bits

= 64 bits

[NAT]

#Q. Consider a memory with access speed of 1GBytes per second. Time required to read 128 bytes from memory is ____ nanoseconds?

$1\ GB \rightarrow 1\ s$

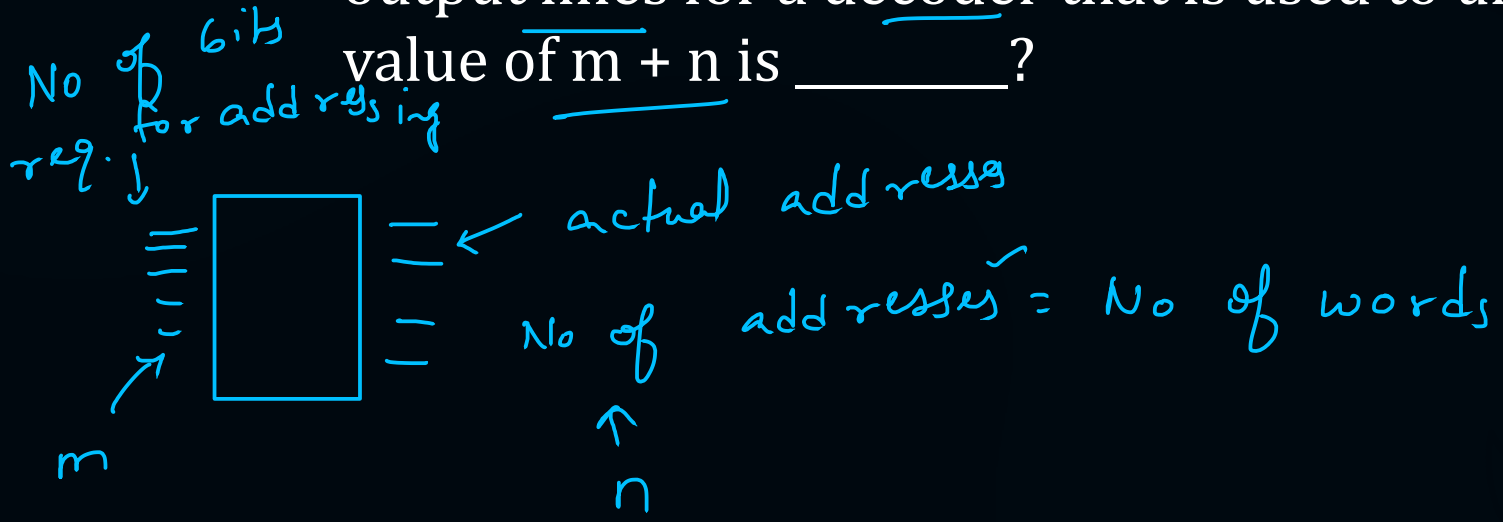
$\Rightarrow 10^9\ B \rightarrow 1\ s$

$\Rightarrow 1\ B \rightarrow \frac{1}{10^9}\ s$

$\Rightarrow 128\ B \rightarrow \frac{128}{10^9}\ s = \frac{128 \times 10^9\ ns}{10^9} = 128\ ns$

[MCQ]

#Q. Consider a memory with a maximum size of 16Kbytes. The memory is word-addressable with the word size of 64 bits. If there are m input lines n output lines for a decoder that is used to uniquely address this memory then value of m + n is _____?



Memory size = 2^{14} B

1 word size = 64 bits = 8 B

No of words = $\frac{2^{14}\ B}{2^3\ B}$

$= 2^{11}$

$n = 2048$

$m = \log_2(2^{11}) = 11$

$m + n = 11 + 2048 = 2059$

THANK - YOU